State Vocational Rehabilitation Program Delivery and Hourly Wages:

A Multilevel Analysis

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Introduction

Background

There is growing attention on the importance of increasing the employment participation of individuals with intellectual and developmental disabilities (IDD). The Rehabilitation Act has become an influential driver of employment outcomes for people with IDD by providing formula grants to states for vocational rehabilitation (VR) programs. Due to the state-by-state implementation of VR services, state and program contextual factors are important to understand when evaluating VR outcomes.

Research Questions

1. How does VR program delivery moderate the average wages of people with IDD?
2. Does VR program delivery affect people with different impairments equally?

Method

Data and Sample

- **Level-1 Data**: The RSA-911 case closure dataset is a public access database from FY11 and provides individual level information on every person who participated in the vocational rehabilitation (VR) program.
- **Level-2 Data**: The RSA Annual Review Report for FY11 provides VR system results for each state. U.S. Dept. of Labor – Wage and Hour Division state non-farm employment minimum wage law dataset.
- **Sample**: Includes all people with IDD that received VR services who were employed and achieved a case closure in FY11 with the sample size, N = 17,888.

Variables

Level-1	Demographic Impairment Sig. Disability	Public benefits usage Job acquisition services Hours at closure
Level-2	Pct. of State VR outcomes in integrated setting State minimum wage – Large businesses (FLSA) State minimum wage – Small businesses	

Data Analysis

- **HLM**: A structured regression approach was used to explain the nested nature of the data –individual VR within a state VR system. Three-step modeling process was applied, building a model upon the previous one.
- **The Unconditional Model**: State nesting was found to have a significant effect on the outcome, t (47)= 77.82, p<.001.

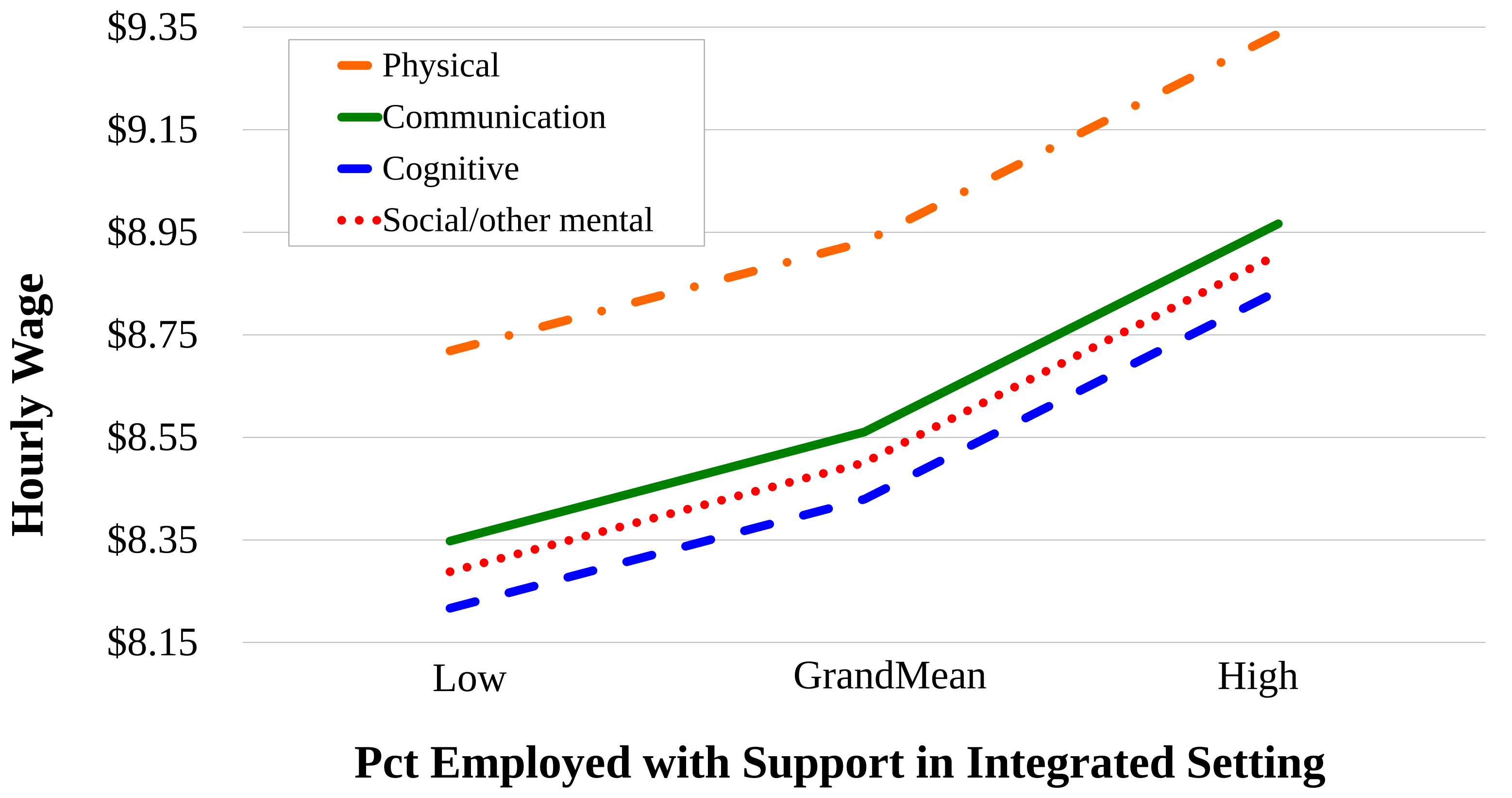
Method

Data Analysis

- **The Intercept Model**: The random intercept model was fitted to the dollars earned per hour with level-1 and level-2 predictors. The slopes are not allowed to vary across the states meaning that at level-2 the slopes are the fixed effects in the model.
- **The Slope-as-Outcome Model**: The important feature of this model is the slopes are allowed to vary across the states at level-2 to find how much the intercepts and the slopes vary. In other words, this model allows investigating the extent to which states differ from each other.
- The level-2 predictors as indicator variables explain random variation in the intercept and slopes and these variables are grand mean centered.

Results

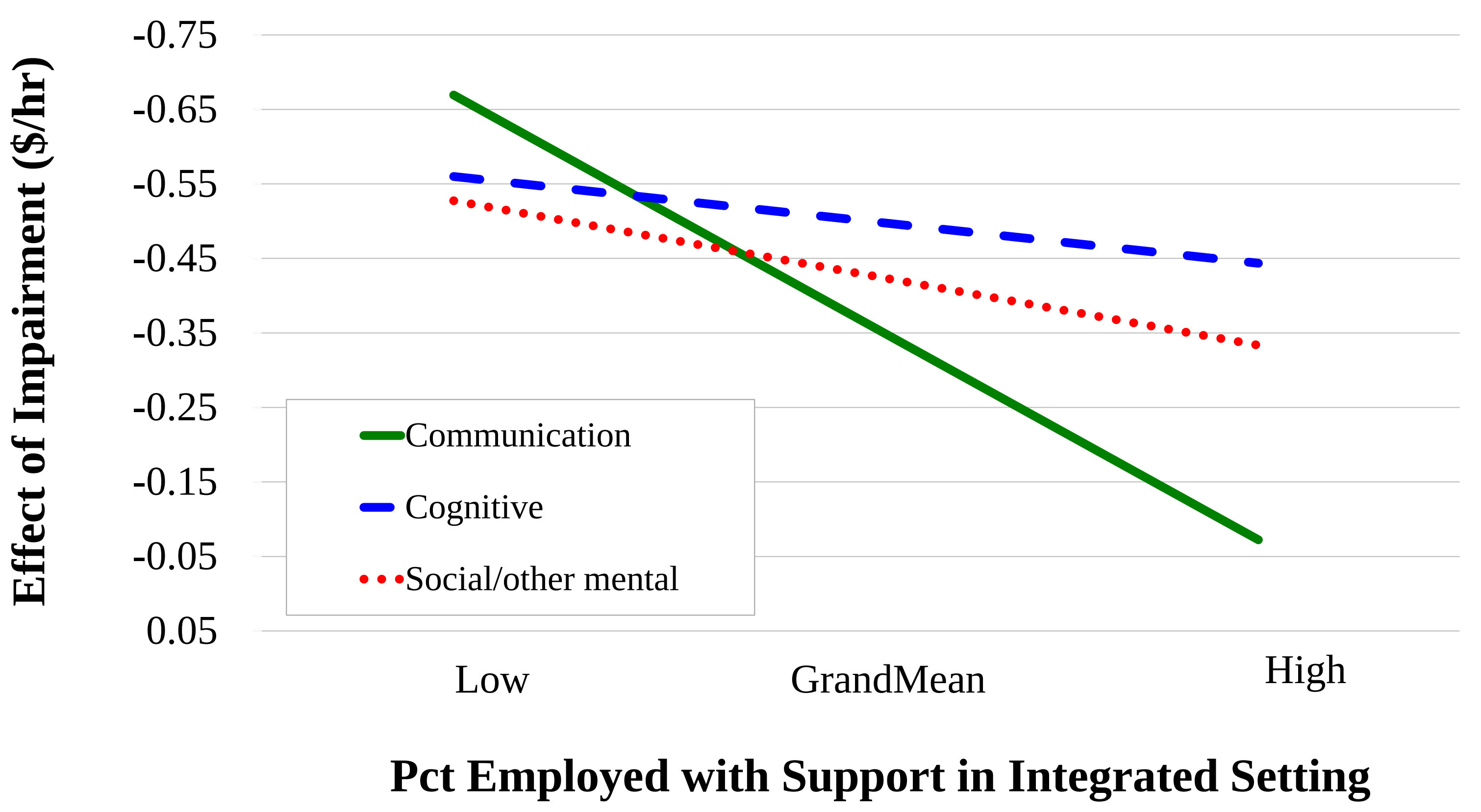
Figure 1 - Intercept Model: Effect of State VR Program on Hrly Wage



- **Figure 1**: Comparison impairments have negative effects on hourly wage and these effects are differential. People with physical impairments have significantly higher hourly wages than those with communication, cognitive, or social/mental impairments.
- **Figure 2**: Note that the physical impairment is the reference group. The average negative effect of impairment on hourly wage was moderated by the state program. Specifically, compared to the reference group, the negative impairment effects were reduced in states with VR agencies with higher % of state program support. The negative effects were exacerbated in states state with lower % of state program support. The positive moderating effects of % of state program support is not distributed equally across people with IDD with different impairments.

Results

Figure 2 - Slope-as-Outcome Model: Inequity of VR Program Effects



In other words, the level of state programmatic benefit on hourly wages is experienced differently based on ones impairment. As shown in the graph, on average those with communication impairments experience the greatest benefits, where the average negative effects of having a communication impairment on hourly wage is nearly **erased** in states with higher % of state program support. Those with cognitive and social/mental impairments also averaged reductions in the negative effects in states with % of state program support, however, these reductions were less dramatic compared to the communication impairment.

Discussion

- VR services are critical supports to many people with IDD to enter and remain in the workforce. In evaluating program outcomes, it is clear that state program delivery matters. It would benefit state VR programs to assess their level of integrated support to those employed and adjust to provide greater levels, since it relates to higher wages.
- Additionally, attention should be paid to those with different impairments. The positive effects of VR program delivery are not equally distributed across individuals. Greater attention to the support needs of people with cognitive and social/other mental impairments is recommended.

References

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